0.0.1 Density of Rationals

Lemma 0.0.1. For any real numbers $a < b$, there exists a rational number $p \in \mathbb{Q}$ such that $a < p < b$.

Proof. If a is rational, then take $N \in \mathbb{N}$ such that

$$\frac{1}{N} < b - a$$

Then, $a + \frac{1}{N}$ is a rational number that we desired. Suppose a is irrational. Define a map as:

$$\lceil \cdot \rceil : \mathbb{R} \to \mathbb{Z} : x \mapsto \min\{n \in \mathbb{Z} \mid n \geq x\}$$

That is, $\lceil x \rceil$ is the smallest integer greater than or equal to x. And, for any $x \in \mathbb{R}$,

$$x \leq \lceil x \rceil \leq x + 1$$

By Archimedean property, there exists $N \in \mathbb{N}$ such that

$$a = \frac{Na}{N} \overset{(*)}{\leq} \frac{\lceil Na \rceil}{N} \leq \frac{Na + 1}{N} = a + \frac{1}{N} < b$$

Since a is irrational, the equality of $(*)$ cannot hold. Now, $\frac{\lceil Na \rceil}{N}$ is a rational number that we desired. □

Corollary 0.0.2. For any real numbers $a < b$, there exists an irrational number $c \in \mathbb{R} \setminus \mathbb{Q}$ such that $a < c < b$.

Proof. By the Lemma, there exists a rational nubmer $p \in \mathbb{Q}$ such that

$$a < p < b$$

By Archimedean property, there exists $N \in \mathbb{N}$ such that

$$a < p < p + \frac{1}{N} < b$$

Now,

$$a < p < p + \frac{1}{2N} < p + \frac{\sqrt{2}}{2N} < p + \frac{1}{N} < b$$

□

0.0.2 Dirichlet function

Theorem 0.0.3. Define a real-valued function as:

$$f : \mathbb{R} \to \mathbb{R} : x \mapsto \begin{cases} 1 & x \in \mathbb{Q} \\ 0 & x \notin \mathbb{Q} \end{cases}$$

Then, f is nowhere continuous.

Proof. Let $x \in \mathbb{R}$ be given. To show f is not continuous at x using contradiction, suppose f is continuous at x. By definition of continuous, there exists $\delta > 0$ such that

$$|x - t| < \delta \implies |f(x) - f(t)| < 1$$

But, the Lemma gives that there exists a rational number p and an irrational number q in $(x - \delta, x + \delta)$.
Now, regardless of x is rational or irrational, we can choose $t = p$ or $t = q$ to make $|f(x) - f(t)| = 1$, contradiction. □